

Intro to Rational Expressions

Fractions and Exponents Review

Fractions Review

Adding and Subtracting Fractions

Always find a common denominator when adding or subtracting fractions!

$$\text{a) } \frac{1 \times 4}{2 \times 4} + \frac{1}{8}$$

$$= \frac{4}{8} + \frac{1}{8}$$

$$= \frac{5}{8} \text{ (keep denominator)}$$

$$\text{b) } \frac{2x \times 2}{3y \times 2} - \frac{y \times 3y}{2 \times 3y}$$

common denom = 3y(2)
= 6y

$$= \frac{4x}{6y} - \frac{3y^2}{6y}$$

$$= \frac{4x - 3y^2}{6y}$$

Multiplying and Dividing Fractions

You do NOT need a common denominator when multiplying or dividing fractions!

a) $\frac{3}{2} \cdot \frac{4}{5}$

$$= \frac{(3)(4)}{(2)(5)}$$

$$= \frac{12}{10}$$

$$= \frac{6}{5}$$

b) $\frac{2}{3} \div \frac{4}{3}$

*Flip second
* fraction and *
multiply*

$$= \frac{2}{3} \times \frac{3}{4}$$

$$= \frac{6}{12}$$

$$= \frac{1}{2}$$

Rule: We can NEVER have a fraction with a denominator of 0. Why?

eg. $\frac{6}{2} = 3$

why? because $2 \times 3 = 6$

$$\left\{ \frac{6}{0} = ? \right.$$

$\left\{ \begin{array}{l} \frac{6}{0} = ? \\ \text{This means } 0 \times \underline{\quad} = 6 \end{array} \right.$

Rule: Cross multiplication of fractions only happens when..... *There is an = sign between two fractions.*

eg. $\frac{2}{3} = \frac{x}{5}$

$$5(2) = 3x$$

$$10 = 3x$$

$$x = \frac{10}{3}$$

Rule: We can cancel out ONLY when multiplying fractions

You can cancel factors.

$$\text{eg. } \frac{\cancel{x}(x+1)}{\cancel{x}} = x+1$$

$$\text{eg. } \frac{3\cancel{x}(x+3)}{2\cancel{4}\cancel{x}} = \frac{3x+9}{2}$$

These are examples of when cancelling is allowed.

Rule: We can NOT cancel out when adding or subtracting fractions

$$\text{eg. } \frac{\cancel{x}+8}{\cancel{x}} \quad \underline{\text{NO!}} \quad \text{Don't do this!}$$

$$\text{eg. } \frac{2x+3}{4x} \quad \text{No cancelling!}$$

Name	Rule	Examples
Adding and Subtracting Monomials	COMBINE LIKE TERMS! (do not change common variables and exponents)	$3x^2y + 2x^2y = 5x^2y$
Product Rule	$x^a \cdot x^b = x^{a+b}$	$(-2x^2y)(3x^3y^2) = -6x^5y^3$
Quotient Rule	$\frac{x^a}{x^b} = x^{a-b}$	$\frac{28x^5}{42x} = \frac{2x^4}{3}$
Power of a Power Rule	$(x^a)^b = x^{a \times b}$	$(-2x^3)^2 = 4x^6$
Negative Exponent Rule	$x^{-a} = \frac{1}{x^a}$	$\frac{4x^2}{8x^5} = \frac{1}{2x^3}$
Exponent of Zero	$x^0 = 1$	$87^0 = 1$

Simplify the following rational expressions using exponent laws

a)
$$\frac{12k^2m^8}{4k^5m^5} = 3k^{-3}m^3$$

$$= \frac{3m^3}{k^3}$$

b)

$$\frac{5c^3d \cdot 4c^2d^2}{(2c^2d)^2} = \frac{20c^5d^3}{4c^4d^2} = 5cd$$

c)

$$\frac{(3xy)^3}{9x^4y^4} = \frac{27x^3y^3}{9x^4y^4} = 3x^{-1}y^{-1} = \frac{3}{xy}$$

d)

$$\frac{(2z^3)^{-2}}{w^5 z^2} = \frac{2^{-2} z^{-6}}{w^5 z^2}$$
$$= \frac{1}{2^2 w^5 z^2 z^6}$$
$$= \frac{1}{4w^5 z^8}$$

e)

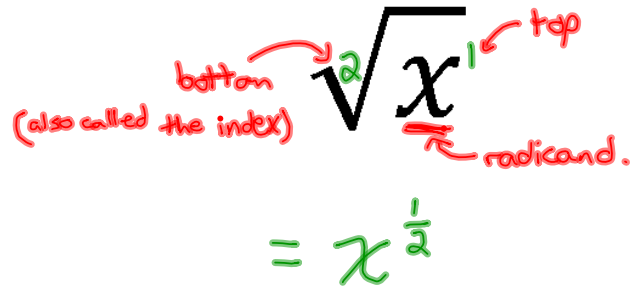
$$\frac{(x^{-4})^5 x^3}{3x^{-1}} = \frac{x^{-20} x^3}{3x^{-1}}$$
$$= \frac{x^{-17}}{3x^{-1}}$$
$$= \frac{1}{3x^{-1} x^{17}}$$
$$= \frac{1}{3x^{16}}$$

Combining fractions and exponents:

$$\begin{aligned}\text{ex. } & \frac{3x^{\cancel{1}}}{2\cancel{x}} + \frac{4y^{\cancel{3}}}{3\cancel{y}} \\ &= \frac{3x^{\times 3}}{2^{\times 3}} + \frac{4y^{\cancel{3} \times 2}}{3^{\times 2}} \\ &= \frac{9x}{6} + \frac{8y^3}{6} \\ &= \frac{9x+8y^3}{6}\end{aligned}$$

COMPLETE WORKSHEET

Intro to Rational exponents (Fractions):



A diagram showing the components of a square root expression. The expression is $\sqrt{x}$. The number 2 (the index) is labeled "bottom (also called the index)" with a red arrow. The variable x is labeled "top" with a red arrow. The entire expression under the radical is labeled "radicand" with a red arrow. Below the expression, the equivalent rational exponent form is shown: $= x^{\frac{1}{2}}$.

Powers with a rational exponent of the form $\frac{1}{n}$

A power involving a rational exponent with numerator 1 and denominator n can be interpreted as the n th root of the base:

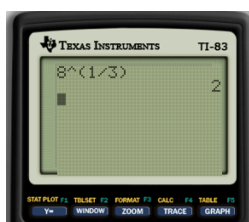
$$b^{\frac{1}{n}} = \sqrt[n]{b}$$

Powers with a rational exponent of the form $\frac{1}{n}$

Example 1: Evaluate each of the following

a) $8^{\frac{1}{3}}$

$$= 2$$



b) $\sqrt[5]{-32}$

$$= (-32)^{\frac{1}{5}}$$

$$= -2$$

c) $-16^{\frac{1}{4}}$

$$= -2$$

d) $\sqrt[4]{\frac{16}{81}}$

$$= \left(\frac{16}{81}\right)^{\frac{1}{4}}$$

$$= \frac{(16)^{\frac{1}{4}}}{(81)^{\frac{1}{4}}}$$

$$= \frac{2}{3}$$

e) $(-27)^{-\frac{1}{3}}$

$$= \frac{1}{(-27)^{\frac{1}{3}}}$$

$$= \frac{1}{-3}$$

$$= -\frac{1}{3}$$

Powers with a rational exponent of the form $\frac{m}{n}$

You can evaluate a power involving a rational exponent with numerator m and denominator n by taking the n th root of the base raised to the exponent m :

$$b^{\frac{m}{n}} = (\sqrt[n]{b})^m = \sqrt[n]{b^m}$$

Powers with a rational exponent of the form $\frac{m}{n}$

Example 2: Simplify each of the following powers

a)

$$\sqrt[5]{y^2}$$

$$= y^{\frac{2}{5}}$$

b)

$$\sqrt[3]{x}$$

$$= x^{\frac{1}{3}}$$

$$\begin{aligned} \text{c) } & \sqrt{a^{-3}b^{\frac{4}{3}}} \\ &= (a^{-3}b^{\frac{4}{3}})^{1/2} \\ &= a^{-3/2}b^{4/6} \\ &= \frac{b^{2/3}}{a^{3/2}} \end{aligned}$$

$$\begin{aligned} \text{d) } & \sqrt[4]{x^3y^2} \\ &= (x^3y^2)^{1/4} \\ &= x^{3/4}y^{2/4} \\ &= x^{3/4}y^{1/2} \end{aligned}$$

$$\begin{aligned} \text{e) } & \frac{\sqrt[3]{x^2yy^2}}{x^3} \\ &= \frac{(x^2y)^{1/3}y^2}{x^3} \\ &= \frac{x^{2/3}y^{1/3}y^{6/3}}{x^3} \\ &= x^{-7/3}y^{7/3} \\ &= \frac{y^{7/3}}{x^{7/3}} \end{aligned}$$

Example 3: Evaluate each of the following

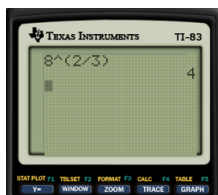
a)

$$8^{\frac{2}{3}}$$

$$= (8^{\frac{1}{3}})^2$$

$$= (2)^2$$

$$= 4$$



b)

$$81^{\frac{5}{4}}$$

$$= (81^{\frac{1}{4}})^5$$

$$= (3)^5$$

$$= 243$$

c)

$$\left(\frac{49}{81}\right)^{-\frac{3}{2}}$$

$$= \left(\frac{81}{49}\right)^{3/2}$$

$$= \frac{(81)^{3/2}}{(49)^{3/2}}$$

$$= \frac{729}{343}$$

If you have a power with a negative exponent and a rational base, just flip the base and make the exponent positive.

Apply Exponent Rules

Example 4: Simplify and express answer using only positive exponents

a)

$$\frac{\left(x^{\frac{2}{3}}\right)\left(x^{\frac{2}{3}}\right)}{\left(x^{\frac{1}{3}}\right)} = \frac{x^{\frac{4}{3}}}{x^{\frac{1}{3}}} = x^{\frac{3}{3}} = x$$

b)

$$\begin{aligned} &\left(y^{\frac{1}{4}}\right)^2 \times \left(y^{-\frac{1}{3}}\right)^2 \\ &= y^{\frac{1}{2}} \times y^{-\frac{2}{3}} \\ &= y^{\frac{3}{6}} \times y^{-\frac{4}{6}} \\ &= y^{-\frac{1}{6}} \\ &= \frac{1}{y^{\frac{1}{6}}} \end{aligned}$$

c)

$$\left(5x^{\frac{1}{2}}\right)^2 \times 4x^{-\frac{1}{2}}$$

$$= 25x \cdot 4x^{-\frac{1}{2}}$$

$$= 25x^{\frac{2}{2}} \cdot 4x^{-\frac{1}{2}}$$

$$= 100x^{\frac{1}{2}}$$

d)

$$\frac{(m^{-2})^3 \sqrt{m^4}}{m \sqrt{pq^{-3}}}$$

$$= \frac{m^{-6} \cdot m^{\frac{4}{2}}}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-6} \cdot m^2}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-4}}{m p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{m^{-5}}{p^{\frac{1}{2}} q^{-\frac{3}{2}}}$$

$$= \frac{q^{\frac{3}{2}}}{m^5 p^{\frac{1}{2}}}$$

e)

$$\frac{(x^2)^{-4} \sqrt[5]{y^3}}{y \sqrt{x^{-2} y}}$$

$$= \frac{x^{-8} y^{3/5}}{y x^{-1} y^{1/2}}$$

$$= \frac{x^{-7} y^{3/5}}{y^{3/2}}$$

$$= \frac{x^{-7} y^{6/10}}{y^{15/10}}$$

$$= x^{-7} y^{-9/10}$$

$$= \frac{1}{x^7 y^{9/10}}$$

Complete Worksheet

2.1/2.2 Restricting, Simplifying, Multiplying, and Dividing Rational Expressions

Lesson Outline:

Part 1: Stating restrictions

Part 2: Simplifying rational expressions

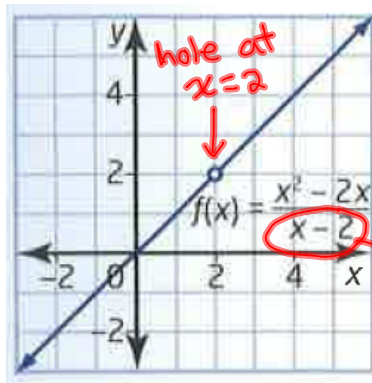
Part 3: Multiplying rational expressions

Part 4: Dividing rational expressions

What is a rational expression?

Rational expression: the quotient of two polynomials, $\frac{p(x)}{q(x)}$, where $q(x) \neq 0$.

Example of a graph of a rational expression:



The open circle is used to represent a hole in the graph. This corresponds to any restrictions on the variable (denominator can't be 0).

$$x - 2 \neq 0$$
$$x \neq 2$$

Stating Restrictions

Note: rational expressions must be checked for restrictions by determining where the denominator is equal to **zero**. These restrictions must be stated when the expression is simplified.

bottom of a fraction can
NOT = 0.

Example 1: State the restrictions for the following rational expressions

a)

$$\frac{x+2}{x-2}$$

$$x-2 \neq 0$$
$$x \neq 2$$

$$x \neq 2$$

b)

$$\frac{x+2}{(x-3)(x+4)}$$

$$x-3 \neq 0$$
$$x \neq 3$$

$$x+4 \neq 0$$
$$x \neq -4$$

$$x \neq 3, -4$$

c)

$$\frac{5}{x(x+3)}$$

$$x \neq 0$$

$$x+3 \neq 0$$
$$x \neq -3$$

$$x \neq 0, -3$$

Rule: We can cancel out **ONLY** when multiplying fractions

You can cancel factors.

e.g. $\frac{\cancel{x}(x+1)}{\cancel{x}} = x+1$

e.g. $\frac{3\cancel{x}(x+3)}{2\cancel{4x}} = \frac{3x+9}{2}$

These are examples of when cancelling is allowed.

Rule: We can **NOT** cancel out when adding or subtracting fractions

e.g. $\frac{\cancel{x}+8}{\cancel{x}}$ **NO!** Don't do this!

e.g. $\frac{2x+3}{4x}$ No cancelling!

Simplifying Rational Expressions

Example 2: Simplifying each expression and determine any restrictions on the variable.

a) $\frac{3x^2}{y\cancel{x}}$, $y \neq 0$
 $x \neq 0$

$$= \frac{3\cancel{x}}{y} , \quad y \neq 0$$
$$x \neq 0$$

b) $\frac{x-3}{x^2+3x-18}$

Note: factor where possible and then state restrictions before cancelling factors.

$$= \frac{\cancel{x-3}}{(x+6)(\cancel{x-3})} , \quad x \neq -6, 3$$

$$= \frac{1}{x+6} , \quad x \neq -6, 3$$

c)

$$\frac{x^2 + 10x + 21}{x + 3} \quad , x \neq -3$$

$$= \frac{(x+7)\cancel{(x+3)}}{\cancel{x+3}} \quad , x \neq -3$$

$$= x + 7 \quad , x \neq -3$$

d)

$$\frac{x + 1}{x^2 + 3x + 2}$$

$$= \frac{\cancel{x+1}}{(x+2)\cancel{(x+1)}} \quad , x \neq -2, -1$$

$$= \frac{1}{x+2}$$

e)

$$\frac{x^2 - 9}{x^2 + 7x + 12} \leftarrow \text{difference of squares: } a^2 - b^2 = (a-b)(a+b)$$

$$= \frac{(x-3)\cancel{(x+3)}}{(x+4)\cancel{(x+3)}} \quad , x \neq -4, -3$$

$$= \frac{x-3}{x+4} \quad , x \neq -4, -3$$

f)

$$\frac{6x^2 - 7x - 5 *}{3x^2 + x - 10 *}$$

factor numerator

$$\begin{aligned} &6x^2 - 10x + 3x - 5 \\ &= (6x^2 - 10x) + (3x - 5) \\ &= 2x(3x - 5) + 1(3x - 5) \\ &= (2x+1)(3x-5) \end{aligned}$$

factor denominator

$$\begin{aligned} &3x^2 + x - 10 \\ &= 3x^2 + 6x - 5x - 10 \\ &= (3x^2 + 6x) + (-5x - 10) \\ &= 3x(x+2) - 5(x+2) \\ &= (x+2)(3x-5) \end{aligned}$$

$$= \frac{(2x+1)\cancel{(3x-5)}}{(x+2)\cancel{(3x-5)}} \quad , x \neq -2, \frac{5}{3}$$

$$= \frac{2x+1}{x+2} \quad , x \neq -2, \frac{5}{3}$$

Multiplying Rational Expressions

a)

$$\frac{2\cancel{4}x^{\cancel{2}^1}}{1\cancel{3}x} \cdot \frac{4\cancel{12}x^{\cancel{2}^2}}{1\cancel{2}x} ; x \neq 0$$

1. factor where possible
2. cancel common factors
3. multiply numerators and denominators
4. state restrictions (throughout process)

$$= 8x^3 ; x \neq 0$$

b)

$$\frac{4x+24}{x^2+8x} \cdot \frac{12x^2}{3x+18}$$

$$= \frac{4(\cancel{x+6})}{x(\cancel{x+8})} \cdot \frac{4\cancel{12}x^{\cancel{2}^1}}{1\cancel{3}(\cancel{x+6})} ; x \neq 0, -8, -6$$

$$= \frac{16x}{x+8} ; x \neq 0, -8, -6$$

c)

$$\frac{x+1}{2x} \cdot \frac{3x}{x^2+4x+3} \quad , x \neq 0$$

$$= \frac{\cancel{x+1}}{2\cancel{x}} \cdot \frac{3\cancel{x}}{(\cancel{x+1})(x+3)} \quad ; x \neq 0, -1, -3$$

$$= \frac{3}{2(x+3)} \quad ; x \neq 0, -1, -3$$

d)

$$\frac{*5x^2-13x+8}{x-7} \cdot \frac{1}{5x-8}$$

$$\begin{aligned} &\text{factor} \\ &= 5x^2 - 8x - 5x + 8 \\ &= (5x^2 - 8x) + (-5x + 8) \\ &= x(5x - 8) - 1(5x - 8) \\ &= (5x - 8)(x - 1) \end{aligned}$$

$$= \frac{(\cancel{5x-8})(x-1)}{x-7} \cdot \frac{1}{\cancel{5x-8}} \quad ; x \neq 7, \frac{8}{5}$$

$$= \frac{x-1}{x-7} \quad ; x \neq 7, \frac{8}{5}$$

Dividing Rational Expressions

a)

$$\frac{10ab^2}{4a} \div \frac{15a^2}{12b^2}$$

*no cross cancelling until
after second fraction has
been flipped*

1. flip second fraction and change to multiplication
2. factor where possible
3. cancel common factors
4. multiply numerators and denominators
5. state restrictions (throughout process)

$$= \frac{10ab^2}{4a} \times \frac{12b^2}{15a^2}$$

$$; a \neq 0$$

$$b \neq 0$$

*because 'b' was
in the denominator of
the original expression*

$$= \frac{\cancel{2} \cancel{12} \cancel{0} ab^4}{\cancel{4} \cancel{15} a^2}$$

$$; a \neq 0$$

$$b \neq 0$$

$$= \frac{2b^4}{a^2} ; a \neq 0$$

$$b \neq 0$$

b)

$$\frac{a^2 + 2a}{3a} \div \frac{5a^2 + 10a}{20a^2}$$

$$= \frac{a^2 + 2a}{3a} \times \frac{20a^2}{5a^2 + 10a} ; a \neq 0$$

$$= \frac{\cancel{a}(\cancel{a} + 2)}{\cancel{3}\cancel{a}} \times \frac{\cancel{4} \cancel{20} a^2}{\cancel{5} a(\cancel{a} + 2)} ; a \neq 0, -2$$

$$= \frac{4a}{3} ; a \neq 0, -2$$

c)

$$\frac{2x^2 - 8x}{x^2 - 3x - 10} \div \frac{4x^2}{x^2 - 9x + 20}$$

$$= \frac{2x^2 - 8x}{x^2 - 3x - 10} \times \frac{x^2 - 9x + 20}{4x^2} ; x \neq 0$$

$$= \frac{\cancel{2}x \cancel{(x-4)}}{\cancel{(x-5)}(x+2)} \times \frac{\cancel{(x-5)}(x-4)}{\cancel{2}4x\cancel{2}} ; x \neq 0, 5, -2, 4$$

$$= \frac{(x-4)^2}{2x(x+2)} ; x \neq 0, 5, -2, 4$$

DO WORKSHEET

2.2 Add and Subtract Rational Expressions

DO IT NOW!

$$LCD = 30$$

$$\text{a) } \cancel{x}^1 \frac{1}{\cancel{x}^6} + \frac{1 \times 6}{5 \times 6}$$

$$= \frac{5}{30} + \frac{6}{30}$$

$$= \frac{11}{30}$$

Note: the product of the denominators will give a common denominator (but not always the lowest common denominator)

b) Simplify and state restrictions

$$\frac{x^2 - 1}{x^2 - 4} \times \frac{x^2 + 3x - 4}{x^2 + 5x + 4} = \frac{(x-1)\cancel{(x+1)}}{(x-2)(x+2)} \times \frac{\cancel{(x+4)}(x-1)}{\cancel{(x+4)}\cancel{(x+1)}}$$

$$= \frac{(x-1)^2}{(x-2)(x+2)} ; x \neq 2, -2, -4, -1$$

c) $\frac{2x}{3y} + \frac{y}{6x}$ $LCD = 6xy$

$$= \frac{4x^2}{6xy} + \frac{y^2}{6xy} \quad ; \begin{matrix} x \neq 0 \\ y \neq 0 \end{matrix}$$

$$= \frac{4x^2 + y^2}{6xy} \quad ; \begin{matrix} x \neq 0 \\ y \neq 0 \end{matrix}$$

Add and Subtract Rational Expressions With Monomial Denominators

a) $\frac{1}{5x} + \frac{1}{2x}$ $LCD = 10x$

$$\frac{2}{10x} + \frac{5}{10x}$$

$$= \frac{2}{10x} + \frac{5}{10x} \quad ; x \neq 0$$

$$= \frac{7}{10x} \quad ; x \neq 0$$

1. factor denominators if possible
2. get a common denominator
3. re-write expression with a common denominator
4. add/subtract the numerator
(keep denominator the same)
5. simplify where possible
6. state restrictions (throughout process)

b)

$$\frac{ab^2 + 2}{2ab^2} - \frac{b + 2}{2b}$$

$LCD = 2ab^2$

$$= \frac{ab^2 + 2}{2ab^2} - \frac{ab(b+2)}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{ab^2 + 2}{2ab^2} - \frac{ab^2 + 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{ab^2 + 2 - ab^2 - 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{2 - 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{\cancel{2}(1-ab)}{\cancel{2}ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{1-ab}{ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

Add and Subtract Rational Expressions with Polynomial Denominators

a)

$LCD: (x-3)(x+2)$

1. factor denominators if possible
2. get a common denominator
3. re-write expression with a common denominator
4. add/subtract the numerator (keep denominator the same)
5. simplify where possible
6. state restrictions (throughout process)

$$\frac{x+5}{x-3} + \frac{x-7}{x+2}$$

$$= \frac{(x+2)(x+5)}{(x+2)(x-3)} + \frac{(x-7)(x-3)}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

$$= \frac{x^2 + 7x + 10}{(x+2)(x-3)} + \frac{x^2 - 10x + 21}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

$$= \frac{2x^2 - 3x + 31}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

b)

$$\frac{x+9}{x^2+2x-48} - \frac{x-9}{x^2-x-30}$$

$$LCD = (x-6)(x+5)(x+8)$$

$$= \frac{\overset{(x+5)}{\cancel{(x+5)}} \overset{(x+8)}{\cancel{(x+8)}} (x+9)}{\overset{(x+5)}{\cancel{(x+5)}} \overset{(x+8)}{\cancel{(x+8)}} (x-6)} - \frac{\overset{(x+8)}{\cancel{(x+8)}} (x-9)}{(x-6) \overset{(x+5)}{\cancel{(x+5)}} \overset{(x+8)}{\cancel{(x+8)}}$$

$$= \frac{(x+5)(x+9)}{(x+5)(x+8)(x-6)} - \frac{(x-9)(x+8)}{(x+5)(x+8)(x-6)}$$

$$= \frac{x^2+14x+45 - (x^2-1x-72)}{(x+5)(x+8)(x-6)}$$

$$= \frac{15x+117}{(x+5)(x+8)(x-6)} \quad ; x \neq -5, -8, 6$$

c)

$$\frac{x-2}{x+2} + \frac{x+10}{x^2+6x+8}$$

$$LCD = (x+2)(x+4)$$

$$= \frac{\overset{(x+4)}{\cancel{(x+4)}} \overset{(x+2)}{\cancel{(x+2)}} (x-2)}{\overset{(x+4)}{\cancel{(x+4)}} \overset{(x+2)}{\cancel{(x+2)}}} + \frac{x+10}{(x+2)(x+4)}$$

$$= \frac{(x+4)(x-2)}{(x+4)(x+2)} + \frac{x+10}{(x+4)(x+2)}$$

$$= \frac{x^2+2x-8+x+10}{(x+4)(x+2)}$$

$$= \frac{x^2+3x+2}{(x+4)(x+2)} \quad \leftarrow \text{factor}$$

$$= \frac{\cancel{(x+2)}(x+1)}{(x+4)\cancel{(x+2)}}$$

$$= \frac{x+1}{x+4} \quad ; x \neq -4, -2$$

d)

$$\frac{2x}{x-1} - \frac{x+2}{x^2+3x-4}$$

$$LD = (x+4)(x-1)$$

$$= \frac{(x+4)2x}{(x+4)(x-1)} - \frac{x+2}{(x+4)(x-1)}$$

$$= \frac{2x(x+4)}{(x+4)(x-1)} - \frac{x+2}{(x+4)(x-1)}$$

$$= \frac{2x^2+8x-x-2}{(x+4)(x-1)}$$

$$= \frac{2x^2+7x-2}{(x+4)(x-1)} ; x \neq -4, 1$$

e)

$$\frac{a+1}{5-2a} - \frac{a-4}{2a-5}$$

Binomial expressions can differ by a factor of -1. Factor -1 from one of the denominators to identify the common denominator. Then simplify each expression and state the restrictions.

$$= \frac{a+1}{-1(-5+2a)} - \frac{a-4}{2a-5}$$

$$= \frac{(-1)a+1}{(-1)(-5+2a)} - \frac{a-4}{2a-5}$$

$$= \frac{-1(a+1)}{2a-5} - \frac{a-4}{2a-5}$$

$$= \frac{-1(a+1) - (a-4)}{2a-5}$$

$$= \frac{-a-1-a+4}{2a-5}$$

$$= \frac{-2a+3}{2a-5} ; a \neq \frac{5}{2}$$